Q18. [redacted] Part I

① I. Feature map and kernel function

Feature map,

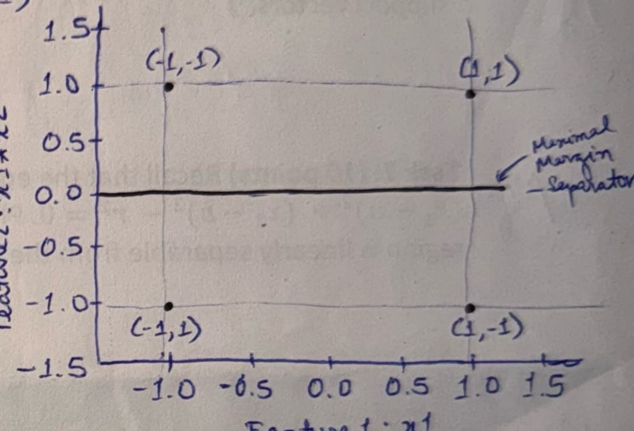
$$\phi(x) = \begin{bmatrix} x_1 \\ x_1 x_2 \end{bmatrix},$$

The corresponding kernel function is

$$K(x,z) = \phi(x)^T \phi(z) = x_1 z_1 + (x_1 x_2)(z_1 z_2) = x_1 z_1 (1 + x_2 z_2)$$

2. Mapped Data Points

Original Input	Label	Mapped $(x_1, x_1 x_2)$
$(-1,-1)$	-1	$(-1,+1)$
$(-1,+1)$	$+1$	$(-1,-1)$
$(+1,-1)$	$+1$	$(+1,-1)$
$(+1,+1)$	-1	$(+1,+1)$



In this feature space, the points form a square.

- Negatives on the top row ($x_1 x_2 = +1$)
- Positives on the bottom row ($x_1 x_2 = -1$)

3. Maximal-Margin Separator.

The best separating hyperplane is:

$$x_1 x_2 = 0$$

A horizontal line in the mapped space (denoted $v = 0$ if $\phi(x) = (u,v)$).

A valid SVM solution is:

$$w = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad b = 0$$

Decision function:

$$f(x) = w^T \phi(x) + b = -x_1 x_2$$

All 4 points satisfy $y_i (w^T \phi(x_i) + b) = 1$ so all are support vectors.

4. Margin Calculation

$$\text{margin} = \frac{1}{\|w\|} = \frac{1}{\sqrt{0^2 + (-1)^2}} = 1$$

Q18. Part(2) Expand the circle equation:

$$(x_1-a)^2+(x_2-b)^2-r^2=0$$

$$\Rightarrow (x_1^2-2ax_1+a^2)+(x_2^2-2bx_2+b^2)-r^2=0$$

$$\Rightarrow x_1^2+x_2^2-2ax_1-2bx_2+(a^2+b^2-r^2)=0$$

$$\Rightarrow (-2a)x_1+(-2b)x_2+(1)x_1^2+(1)x_2^2+(a^2+b^2-r^2)=0$$

Map to the Feature Space

New feature space where each point (x_1, x_2) is mapped to a new point in a 4-dimensional space.

$$z_1=x_1,\ z_2=x_2,\ z_3=x_1^2,\ z_4=x_2^2$$

Substituting these new coordinates into our rearranged equation:

$$(-2a)z_1+(-2b)z_2+(1)z_3+(1)z_4+(a^2+b^2-r^2)=0$$

Linear Separability

A linear equation defines a **hyperplane**. We can write the equation above in the standard form for a hyperplane, which is $w.z+c=0$, where w is a weight vector and c is a bias term.

In our case:

1) The weight vector is $w=\begin{pmatrix}-2a\\-2b\\1\\1\end{pmatrix}$

2) The feature vector is $z=\begin{pmatrix}z_1\\z_2\\z_3\\z_4\end{pmatrix}=\begin{pmatrix}x_1\\x_2\\x_1^2\\x_2^2\end{pmatrix}$

3) The bias term is $c=a^2+b^2-r^2$

The equation of the circle in the original space has been transformed into the equation of a hyperplane in the 4D feature space. This hyperplane divides the 4D space into two halves:

1. Points inside the circle, where $(x_1-a)^2+(x_2-b)^2-r^2<0$, now satisfy $w.z+c<0$
2. Points outside the circle, where $(x_1-a)^2+(x_2-b)^2-r^2>0$, now satisfy $w.z+c>0$

Therefore, the circular region is **linearly separable** by a single hyperplane in that feature space.